

Smart Cycle Count Scoring

Tech Stack:

Next.js + NestJS + Prisma + Postgres

What We Want

We want a small app that helps warehouse teams **decide which locations (“bins”) to audit first** when checking inventory. Instead of counting everything randomly, we’ll assign each bin a **“risk score”** — higher score means it’s more likely to have mistakes. Then we’ll show a **color-coded warehouse map** (heatmap) where bins with higher scores are highlighted.

From there, a user can:

- See which bins need to be audited,
 - Create a small **audit plan** that lists the bins to check,
 - Record what they counted (pass/fail),
 - And have the system get smarter next time.
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Goals for this MVP

By the end, we should be able to:

- Open a dashboard → see a **heatmap of bins with colors** (low risk = green, high risk = red).
- Click a bin → see its **score and “why”** it got that score.
- Click “Generate Audit Plan” → get a list of **Top N risky bins**.
- Open a simple **audit page** (mobile-friendly) → enter the quantity counted and mark pass/fail.
- Run a “recompute scores” action → heatmap updates.

Core Features

1. Seeded Warehouse Data

- Create a small warehouse with aisles, racks, and ≈30 bins.
- Seed fake **pallet data with products inside** and assign these pallets to bins.
- Simulate the last month of warehouse activity:
 - Pallets being placed (putaway),
 - Picked (removed),
 - Adjusted or moved between bins.

2. Scoring System

Give every bin a score between 0–100 based on a few factors you choose:

- Example factors:
 1. Time since last audited
 2. Number of picks/moves/put-away
 3. Adjustment frequency
 4. *Any other factor you believe could help*
- Save score + factor breakdown (so we can show “why” in the UI).
- Add a button or endpoint to recompute scores.

3. Heatmap Dashboard

- Show bins in a simple grid layout per aisle/rack.
- Color bins based on their risk score (green → yellow → red).
- Clicking a bin opens a drawer/page with:
 - Current score
 - Last audit date
 - List of pallets currently stored in that bin

Show bins in a simple grid layout (per aisle/rack).

4. Audit Plan & Tasks

- Button: “Create Plan → Top N risky bins.”
- Save those bins as tasks with status (PENDING/DONE).
- Simple table view of tasks.

5. Count Flow

- Mobile-friendly page:

- Search or scan bin code
 - Show expected pallets inside
 - Enter counted quantity and mark as audited, pass/fail
 - Save the result and trigger a score recompute.
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Deliverables

- **Running app** (Next.js + NestJS) with:
 - Database schema & seed script
 - Simple scoring service
 - Heatmap page & bin details
 - Audit plan creation & task view
 - Mobile count flow
- **README.md** explaining:
 - How to install, seed, and run the app
 - The scoring factors you used and how you combined them